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A

Project Proposal on

“KeyDash”

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Chapter 1: Introduction

1.1 Background

In today's digital age, typing speed accuracy, and adaptability across languages and coding are essential skills for students, professionals, developers, and casual users alike. A Type Speed and Accuracy Tester application helps users measure and improve their typing efficiency. "KeyDash" is an upgraded web-based application designed to evaluate typing speed in words per minute (WPM) and accuracy while providing a user-friendly interface. It now offers multilanguage support, coding practice, multiplayer typing races, AI-based battle modes, symbol mastery, gamification elements like achievements and streaks, and social media sharing. The project leverages modern web technologies like HTML, Tailwind.css, Supabase (postgreSQL), and React.js to create an interactive and responsive tool.

1.2 Statement of Problem

Existing typing test applications often suffer from outdated designs, limited functionality, and a lack of engaging features. Many fail to provide real-time feedback on essential performance metrics such as words per minute (WPM), typing accuracy, and error counts. They lack advanced features that are crucial for comprehensive skill development, such as multi-language support, coding practice and mastery of symbols and special characters. Additionally, existing platforms rarely include motivational and engagement-driven elements like daily streaks, challenges, achievements, multiplayer, AI-based battle and leaderboards. Furthermore, many current applications have unintuitive interfaces, and provide minimal real-time feedback or detailed analytics, reducing their both educational and professional purposes. Furthermore, most applications do not offer integration with platforms like GitHub for version control and collaboration, nor do they incorporate entertainment elements that could enhance user engagement. KeyDash seeks to address these shortcomings by delivering a modern, interactive,

feature-rich typing platforms that emphasizes real-time performance tracking, multi-language typing, coding, real-time multiplayer typing races, AI-based battle modes, symbol practice, gamification elements, and social sharing, a user-friendly and attractive interface, multiple typing modes, GitHub integration, and an enjoyable interactive experience.

1.3 Objective

The primary objectives of KeyDash are:

- To develop a responsive and interactive typing tester with multilanguage support, coding and symbols typing models, real time multiplayer competitions, AI-based typing battles, and detailed performance tracking using HTML, React, js.
- To provide real-time WPM, gaming features like daily streaks, leaderboards and one-click social media sharing to enhance user engagement.

1.4 Scope of Project

- Professional & Educational Use: Coding typing mode benefits students and developers; professional and casual users can get benefits from structured typing practice.
- Skill Improvement: Users can practice multi-language typing, real-time multiplayer typing races, coding syntax, and symbols to improve typing versatility.
- Analytics & Tracking: Detailed typing history helps users track progress over time.
- Gamification & Motivation: Daily streaks, challenges, achievements, and leaderboards encourage consistent practice.

1.5 Limitations

- Limited to web browsers (no standalone mobile app).
- Real-time multiplayer performance depends on internet stability.
- Language support is currently limited to English and Nepali only

Chapter 2: Literature Review

Typing speed and accuracy have long been recognized as essential skills in both educational and professional environments. Traditional typing platforms, such as 10FastFingers, primarily focus on measuring words per minute(WPM) and basic accuracy. While they provide a quick assessment of typing ability, these platforms often lack personalized feedback and adaptive difficulty, limiting their effectiveness.

Research has shown that accurate typing reduces cognitive load, allowing users to focus more on content creation rather than the mechanics of typing. Karat et al.(1999) highlighted that improving typing accuracy helps minimize errors and reduces mental strain. Similarly, Rogers and Monsell(1995) found that faster typing speeds significantly reduce task completion time in computer-based workflows. Recent studies emphasize the importance of interactive learning tools. Chen et al.(2021) demonstrated that providing real-time feedback and analytics during skill development enhances retention.

Current typing applications also show a catering to specialized typing needs. For example, coding typing modes for programming languages like python , java and javaScript are rarely available. Similarly, typing mastery of symbols, multiplayer races, AI-battle modes and special characters is often neglected, despite being essential for efficient programming, and interactive practice. Additionally, social sharing features, which can promote engagement and recognition of progress, are mostly absent in existing tools.

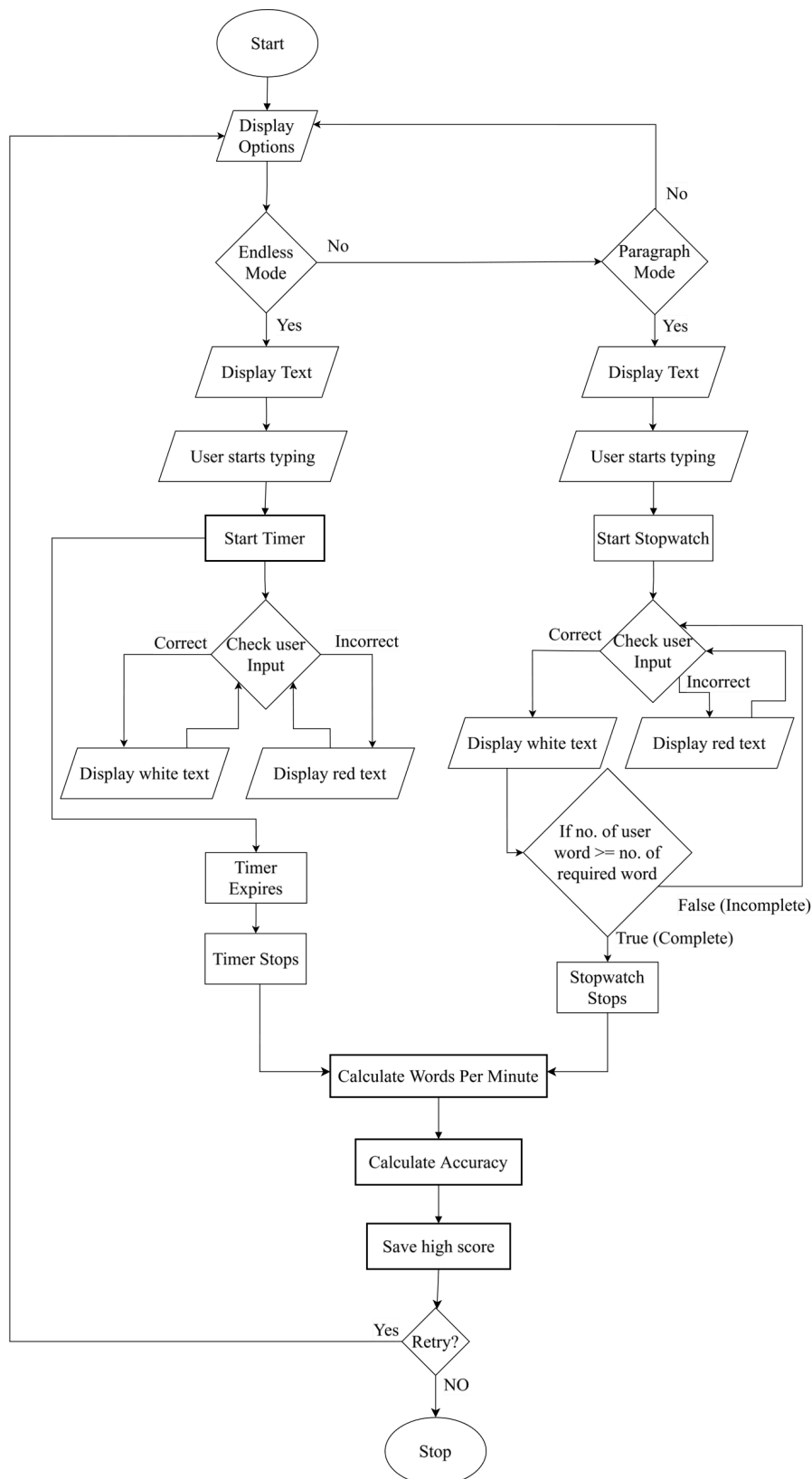
KeyDash aims to address these gaps by providing an interactive and feature-rich typing platform. With multi-language support(English & Nepali), coding and symbols typing modes, real-time feedback, multiplayer typing races and AI based typing modes. KeyDash not only measures typing performance but also encourages consistent improvement and makes the learning experience enjoyable and engaging for users of all levels.

Chapter 3: Methodology

3.1 Approach

- Agile Development: Iterative sprints with GitHub for task tracking.
- Tech Stack:
 - Frontend: React.js (hooks, components), Tailwindcss for styling.
 - Backend: Supabase for authentication, saving history and leaderboard storage.
 - Storage: Supabase storage (Share cards and assets)
 - Security: Row Level Security (RLS)
 - Version Control: GitHub (branches, pull requests).
 - Saving system: Firebase

3.2 Flowchart



3.3 Tools

Tools / Technology	Purpose
HTML	Structure and semantic markup of web pages.
Tailwindcss	Utility-first CSS framework for styling.
React.js	JavaScript library for building user interface.
VS code	Code editor with extensions for development
Git	Version control.
GitHub	Remote repository hosting.
Supabase	Score saving system.
Figma	UI/UX design and prototyping.

Chapter 4: Project Schedule

Gantt Chart:

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7
Research & Planning							
UI Design							
Core Functionality							
Testing & Debugging							
Final Documentation							

Chapter 5: Expected Outcomes

- Typing Performance Metrics:
 - Accurately measure user's typing speed (WPM) and error rates with real-time feedback.
- A fully functional web-based typing tester built with React.js, featuring:
 - One-click social media sharing of results and AI-based typing battles.
 - Multi-language typing (English and Nepali), multiplayer typing races, and coding typing mode(python, c++) .
 - Symbols typing mode, daily streaks ,challenges and achievements.
 - Real-time WPM/accuracy dashboards.
- Modular Design:
 - Lay groundwork for future features (e.g., multiplayer mode, AI-driven feedback) via a scalable React.js architecture.
- Saving System:
 - Save detailed typing history and analytics for long-term improvement.

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